

RAPTOR-HUNTY

Stealth Recon & Vulnerability Assessment Framework

Automated penetration testing assistant built on Metasploit Framework with multi-language support, noise-level control, and structured reporting.

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Platform	Kali Linux / Metasploit Framework
Classification	Educational Use — Ethical Hacking Lab

1. Project Overview

Raptor-Hunty is an automated penetration testing assistant designed for controlled lab environments. It wraps Metasploit's msfconsole and Nmap into a guided interactive shell that walks a security analyst through the full recon-to-report pipeline from target selection to structured vulnerability output without requiring manual RC file authoring.

Built as part of a real-world ethical hacking lab project, this tool demonstrates how to automate early phase network reconnaissance tasks while keeping full control over scan aggression levels, OS targeting, and output formatting.

1.1 What Problem Does It Solve?

In a typical penetration testing engagement, a junior analyst must manually chain together Nmap, Metasploit's database commands, and report writing a process that is error-prone and time-consuming. Raptor-Hunty automates this chain, reducing setup time and ensuring consistent, reproducible results across every lab session.

1.2 Core Capabilities

Feature	Description
Multi-language UI	Interactive prompts available in Spanish and English, selectable at runtime.
OS-aware scanning	Adjusts scan logic and Nmap flags based on the target OS (Windows, Linux, macOS, Android).
Noise level control	Three modes — Noisy (-T4), Semi-noisy (-T3), Quiet (-T2 -sS -Pn) — to match engagement rules.
Vuln detection	Nmap --script vuln + vulners script for CVE correlation on detected services.
Metasploit DB integration	db_nmap populates hosts, services, and vulns into the MSF database automatically.
Module suggestions	Recommends MSF auxiliary modules per detected service (SSH, HTTP, SMB, FTP, RDP).
Auto-report	Saves a structured TXT report to the Desktop after every run. No manual export needed.
Access control	Authorized users whitelist checked before execution. Refuses to run for unlisted users.
Integrity check	SHA-512 hash validation of the RC file prevents tampering between sessions.
Progress bar	Live whiptail gauge displayed during scan execution for user feedback.

2. Architecture & How It Works

Raptor-Hunty is composed of two main files: a Bash entry point (run.sh) that handles user interaction and environment setup, and a Metasploit RC file (lab_menu.rc) with embedded Ruby that performs the actual scanning and reporting.

2.1 Execution Flow

1. run.sh starts and calls check_access() to verify the current user is in the authorized list.
2. check_files() validates the SHA-512 hash of lab_menu.rc to ensure file integrity.
3. show_banner() displays the centered ASCII art header.
4. choose_language() lets the user pick Spanish or English.
5. choose_input() collects LHOST, RHOSTS, OS type, and noise level interactively.
6. msfconsole is launched with the -r flag pointing to lab_menu.rc.
7. Inside lab_menu.rc, db_nmap runs the Nmap scan and populates the Metasploit database.
8. A Ruby block reads detected services and prints module suggestions with CVE hints.
9. A second Ruby block writes the structured report file to the Desktop.
10. The progress bar (whiptail gauge) runs in a background process and is killed on completion.

2.2 Project Structure

run.sh	Main entry point (Bash). Handles UI, validation, and launches msfconsole.
lab_menu.rc	Metasploit RC file with embedded Ruby. Performs scanning and reporting.
lab_menu.rc.sha512	SHA-512 integrity hash of lab_menu.rc. Auto-generated on setup.
authorized_users.txt	Whitelist of system usernames permitted to run the tool.

3. Setup & Usage

3.1 Requirements

- Kali Linux (or any distro with Metasploit Framework installed)
- msfconsole, nmap, whiptail, sha512sum (all pre-installed on Kali)
- PostgreSQL running and connected to Metasploit (run: msfdb init)

3.2 Installation

Step 1 — Create the project folder:

```
mkdir -p ~/lab_msf && cd ~/lab_msf
```

Step 2 — Add your system username to the authorized list:

```
echo "$(whoami)" > authorized_users.txt
```

Step 3 — Generate the integrity hash:

```
sha512sum lab_menu.rc > lab_menu.rc.sha512
```

Step 4 — Give execute permission to the entry point:

```
chmod +x run.sh
```

Step 5 — Run the tool:

```
./run.sh
```

3.3 Interactive Session Example

Note: The following shows a typical session in English mode with Semi-noisy scan against a Windows target.

```
1) Espanol  2) English
> 2
```

```
Your local IP (LHOST): 192.168.10.100
Network range (RHOSTS, e.g. 10.0.2.0/24): 192.168.10.0/24
```

```
Choose OS: 1) Windows  2) Linux  3) macOS  4) Android  5) Automatic
> 1
```

```
Choose noise level: 1) Noisy  2) Semi-noisy  3) Quiet
> 2
```

```
[*] Starting...
[#####           ] 45% Raptor is tracking...
```

```
[*] Hosts found:
- 192.168.10.10
- 192.168.10.20
```

```
[*] Services found:
- 192.168.10.10:445/tcp  smb  Microsoft Windows SMB
  -> review: auxiliary/scanner/smb/smb_version
  -> possible CVE: CVE-2017-0144 / MS17-010 if applicable
```

```
[+] Report saved to Desktop: ~/Desktop/raptor_report.txt
```

4. Technical Reference

4.1 Noise Level Reference

The noise level determines how aggressively Nmap scans the target. Choose based on your engagement rules.

Level	Nmap Flags	Use Case
Noisy	-sV -O -A -T4 --script vuln	Full fingerprint, fast. Best for lab use where detection does not matter.
Semi-noisy	-sV -O -T3 --script vuln	Balanced. Default mode for lab sessions. Generates fewer alerts.
Quiet	-sS -Pn -n -T2	Stealth mode. Avoids IDS detection. Slower. Use in sensitive environments.

4.2 Service to Module Mapping

When Metasploit detects a service, lab_menu.rc automatically suggests the relevant auxiliary scanner module and flags any known CVE references.

Service	Suggested MSF Module	CVE Reference
SSH	auxiliary/scanner/ssh/ssh_version	OpenSSH version-dependent
HTTP/HTTPS	auxiliary/scanner/http/http_version	Web server version-dependent
SMB (445)	auxiliary/scanner/smb/smb_version	CVE-2017-0144 (MS17-010) if applicable
FTP	auxiliary/scanner/ftp/ftp_version	FTP version-dependent
RDP (3389)	auxiliary/scanner/rdp/rdp_scanner	NLA configuration review

4.3 Output Files

File	Location	Content
raptor_report.txt	~/Desktop/	Structured summary of hosts, services, and vulnerabilities.
raptor_log.txt	~/Desktop/	Full msfconsole session output for review.

5. Security Design Decisions

Although Raptor-Hunty is a tool for offensive security testing, it incorporates defensive design patterns to prevent misuse, ensure reproducibility, and demonstrate security-conscious development.

5.1 Authorized Users List

Before executing any code, `run.sh` reads `authorized_users.txt` and compares the current system user against the whitelist. If the user is not listed, the script exits immediately with an error. This prevents accidental or unauthorized execution in shared environments.

```
grep -qx "$user" "$AUTH_FILE" || { echo "[!] Unauthorized user"; exit 1; }
```

5.2 SHA-512 Integrity Verification

Before loading `lab_menu.rc` into `msfconsole`, the script verifies its hash against `lab_menu.rc.sha512`. If the file has been modified since the hash was generated — whether by accident or tampering — the script refuses to proceed.

```
sha512sum -c "$HASH_FILE"
```

5.3 No Hardcoded Credentials

All target parameters (LHOST, RHOSTS, OS type, noise level) are collected interactively at runtime and exported as environment variables. Nothing sensitive is stored in the scripts.

5.4 Input Validation

Both LHOST and RHOSTS are validated against strict regex patterns before the scan starts. An invalid IP or CIDR block causes the prompt to loop rather than passing malformed data to Nmap.

```
valid_ip() { [[ $1 =~ ^([0-9]{1,3}\.){3}[0-9]{1,3}$ ]]; }  
valid_cidr() { [[ $1 =~ ^([0-9]{1,3}\.){3}[0-9]{1,3}/[0-9]+$ ]]; }
```

6. Legal Disclaimer

WARNING: This tool is built for authorized lab environments only. Running network scans against systems you do not own or have explicit written permission to test is illegal in most jurisdictions. The author takes no responsibility for misuse. Always operate within the scope of a signed Rules of Engagement document.

7. Author

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GitHub Tags

penetration-testing • metasploit • nmap • bash-scripting • ethical-hacking • network-recon • vulnerability-assessment • kali-linux • cybersecurity • lab-environment • cve-detection • automated-reporting